

16/9/26

MOBILE PROGRAMMING II

SOFTWARE ENGINEERING

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AI USAGE DECLARATION

Through this document, I hereby declare that during the development of this capstone project, an Artificial Intelligence assistant (Gemini) was utilized as a technical, methodological, and learning support tool.

The scope of this tool was strictly limited to resolving technical bottlenecks, expediting code structure creation, and diagnosing configuration errors. The specific use cases are outlined below:

- **Database Reverse Engineering:** Generation of SQL schema code and the Entity-Relationship diagram based on the visual structure of the usuarios, roles, and mensajes_contacto tables configured in phpMyAdmin.
- **Data Mapping (Frontend):** Creation of TypeScript interfaces required for strongly typing incoming data from the PHP backend to the Ionic/Angular application.
- **API Service Structuring:** Design of the ApiService class in Angular utilizing HttpClient to define full CRUD (Create, Read, Update, Delete) operations, ensuring seamless communication with the local Apache server (XAMPP).
- **Git Conflict Resolution:** Step-by-step assistance in identifying and resolving a nested submodule issue, enabling a successful push of the frontend code to the remote GitHub repository.

It is important to emphasize that **all implementations generated or suggested by Artificial Intelligence were manually reviewed, adapted to business logic, and thoroughly tested** in the local environment (localhost) to guarantee proper functionality prior to final integration and version control submission.